

BORN EL-Completion & Bridge Rules:

% Rule for handling existential restrictions (hasShape some X)

subx(X, exists(R, B)) :-

 gci(A, exists(R, B)),

 subx(X, A),

 con(X),

 con(A),

 con(B),

 role(R).

% Rule for handling subclass inheritance through properties

subx(X, B) :-

 gci(exists(R,A), B),

 subx(X, exists(R, Y)),

 subx(Y, A),

 con(X),

 con(Y),

 con(A),

 con(B),

 role(R).

% Rules to avoid empty predicates of entities

con(-).

role(-).

indiv(-).

% Bridge Probabilistic Contexts to EL-Completion

subx(X, 'http://www.ontology.org/traffic/bayesian-speed#EuropeanContext') :-

'http://www.ontology.org/traffic/bayesian-speed#EuropeanContext', con(X).

subx(X, 'http://www.ontology.org/traffic/bayesian-speed#USContext') :-

'http://www.ontology.org/traffic/bayesian-speed#USContext', con(X).

Test 1 (European Context): Querying the probability that a Speed Limit Sign has a circular shape.

- **Query:**

```
query(subx('http://www.ontology.org/traffic/bayesian-speed#SpeedLimitSign',  
exists('http://www.ontology.org/traffic/bayesian-speed#hasShape',  
'http://www.ontology.org/traffic/bayesian-speed#ShapeCircle'))).
```
- **Result: 0.8000** (Calculated perfectly against the 0.8 network weight).

Test 2 (US Context): Querying the probability that a Speed Limit Sign has a rectangular shape.

- **Query:**

```
query(subx('http://www.ontology.org/traffic/bayesian-speed#SpeedLimitSign',  
exists('http://www.ontology.org/traffic/bayesian-speed#hasShape',  
'http://www.ontology.org/traffic/bayesian-speed#ShapeRectangle'))).
```
- **Result: 0.2000** (Calculated perfectly against the 0.2 network weight).

Part 3:

Query Executed:

```
query(sub('ont:ClinicCase', 'ont:UrgentTreatmentCase')).
```


Calculated Result: 0.0189